InternX — Autonomous AI Agent
For Internship Discovery & Application Management

Track: Education & Career Tech
Team Name: Team InternX

Event: DeltaCCE Agentic AI Product Build Sprint
Team Members: 1. Trimil Triliver John
2. Vimal Jimmy
3. Jefin Joseph
Agent Framework: Groq LLM Engine (LLaMA 3.3 70B) + Multi-Step Tool Pipeline

AGENTIC AI PRODUCT BUILD SPRINT — TEAM INTERNX SLIDE 1 OF 6

Slide 2: Problem Statement

College students struggle to secure relevant internships due to high fragmentation across hiring portals, leading to massive manual overhead, missed deadlines, and low response rates.

■ Fragmented Discovery

Opportunities are scattered across dozens of platforms
(LinkedIn, Unstop, Glassdoor, Internshala), making manual
searching exhaustive.

■ Generic Resume Rejections

Students submit identical resumes across companies without customizing for role-specific ATS keywords.

■ Blind Qualification Checks

Students apply to roles without knowing if they satisfy strict
GPA, degree, or skill eligibility criteria.

■ Application Pipeline Chaos

No unified system exists to track application statuses, follow-ups, pitch drafts, and interview rounds.

AGENTIC AI PRODUCT BUILD SPRINT — TEAM INTERNX SLIDE 2 OF 6

Slide 3: Solutions and Key Features

Feature Node Capability & Innovation Impact

■ Autonomous Portal Scraper

Aggregates listings automatically across 6 tech domains (AI/ML, Full Stack, Data Science, Security, UI/UX, Product).

■ 40/30/20/10 ATS Rubric

Calculates objective compatibility scores: 40% Core Skills, 30% Eligibility (GPA/Major), 20% Preferred Tools, 10% Domain Alignment.

✍■ Tailored Application Kit

Auto-generates customized cover letters, pitch notes, and highlights missing skill gaps for every target role.

■■ Persistent Kanban Memory

Tracks application status across Discovered, Eligible, Applied, Interviewing, and
Offer stages with local/SQLite state sync.

■ Resume AI Parser

Extracts candidate skills, education, and GPA automatically from raw resume
text.

AGENTIC AI PRODUCT BUILD SPRINT — TEAM INTERNX SLIDE 3 OF 6

Slide 4: Tech Stack and System Architecture

Layer Technologies Used Role in System

Frontend UI React + TypeScript + Vite
Tailwind CSS + FontAwesome

Responsive student dashboard, live agent execution console, filter pills, and side-by-side application kit modal.

Backend API Python 3.10+ & FastAPI
Uvicorn ASGI Server

High-performance REST API handling profile ingestion, tool invocation, rubric scoring, and database transactions.

AI / Agent Groq SDK (`llama-3.3-70b-versatile`)
Deterministic Fallback Matrix

Sub-second LLM reasoning engine for ATS score
calculation, cover letter synthesis, and skill gap
extraction.

Database SQLite (`internx.db`) + LocalStorage Dual-layer persistence ensuring persistent Kanban
pipeline state across browser restarts.

AGENTIC AI PRODUCT BUILD SPRINT — TEAM INTERNX SLIDE 4 OF 6

Slide 5: Agentic AI Architecture (Core Agent Loop)

The InternX Agentic Execution Pipeline (5 Specialized Nodes):

Node 1: Goal Planning

Decomposes candidate profile (GPA, skills, major, target roles) into actionable sub-tasks for multi-step execution.

Node 2: Scraper Tool

Queries external opportunity feeds across 6 domains (AI/ML, Full Stack, Data Science, Security, UI/UX, Product).
Node 3: ATS Evaluator
Tool

Executes the 40/30/20/10 ATS weighted rubric. Verifies eligibility rules and ranks compatibility scores.

Node 4: App Kit Drafter
Tool

Invokes Groq LLM (`llama-3.3-70b-versatile`) to generate tailored pitch notes, cover
letters, and missing skill advice.

Node 5: Memory Sync Tool

Updates persistent database state (`internx.db` / LocalStorage) and syncs application
Kanban pipeline.

AGENTIC AI PRODUCT BUILD SPRINT — TEAM INTERNX SLIDE 5 OF 6

Slide 6: Conclusion & Vision

■ Key Impact Delivered

• 90% Time Saved: Reduces job discovery and cover letter drafting from hours to 5 seconds.
• Zero Blind Rejections: Objective ATS rubric ensures students only apply to roles where they are eligible.
• Transparent AI Audit: Full JSON trace audit log for verifiable decision transparency.

■ Future Roadmap

• Auto-Apply Bot: Browser automation agent for 1-click submission on external portals.
• Mock Interview Agent: AI interview prep tailored to job description requirements.
• University Portal Integration: Direct API sync with college
placement cells.

Thank You! Team InternX

Trimil Triliver John | Vimal Jimmy | Jefin Joseph

AGENTIC AI PRODUCT BUILD SPRINT — TEAM INTERNX SLIDE 6 OF 6